

Identifying opportunities to improve fixed and mobile plant asset availability at ARM Bokoni

We tackle complex challenges by recognising that intricate systems are made up of simpler components. Our solutions are straightforward, delivering a high return on investment upon implementation.



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Executive Summary

Current Understanding

The sustained **decline in platinum group metal (PGM) prices** has placed considerable pressure on producers to **operate at lower cost while maximizing the effectiveness of both fixed and mobile assets**. For Bokoni Mine, these market dynamics present a critical challenge: maintaining profitability under constrained margins while preparing to offset potential production shortfalls from opencast operations, which may face temporary stoppages. This scenario places an urgent need on the underground operation to ramp up production to planned or nameplate output.

Complication

Without high equipment reliability and optimal asset utilisation, the underground operation may struggle to meet future production expectations. Inefficiencies in asset management - such as unplanned downtime, reactive maintenance, and underutilisation of critical equipment - pose a direct threat to achieving sustainable ramp-up. Moreover, current systems and front-line maintenance teams may lack the necessary discipline, data-driven insights, and execution support to drive sustainable improvements.

Required Solution

Expert Global Consulting Agency (EGCA) proposes a strategic partnership with Bokoni Mine to support its underground ramp-up through a proven methodology focused on asset optimisation and performance improvement. Our approach directly addresses the need for cost-effective production by unlocking the full potential of existing equipment and infrastructure. With extensive experience in helping mining operations improve asset reliability and operational discipline, EGCA brings clarity on which performance levers to focus on—and how to activate them sustainably.

Key elements of our solution include:

- A diagnostic of current equipment reliability and utilisation
- Implementation of proactive maintenance and condition monitoring practices
- Integration of a fit-for-purpose Management Operating System (MOS)
- Coaching and support to drive ownership and execution across operational levels
- Targeted interventions to quickly stabilise asset performance and build sustainable improvement



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EGCA CONSULTING

Expert Global Consulting Agency

Who are we?

EGCA is a world class consulting firm dedicated to providing simple solutions to complex problems and deliver tangible return on investment for our partners.

Operating
Since
2007

30+
successful
consulting
clients

\$100m+
in bottom-line
client results

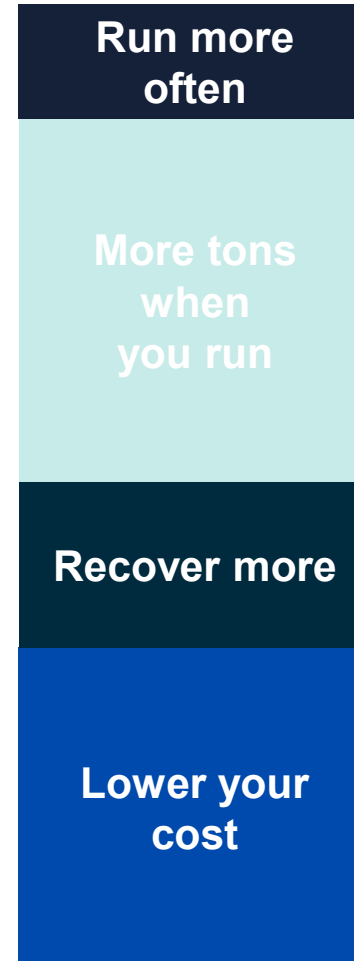
Partnerships with:

XMPRO

XMPPro is a No Code Digital Twin Composition
Platform that accelerates Operational and
Situational Awareness

More importantly – We deliver results and not presentations

Key ROI Value levers



Illustrative

- 1 Reduce duration of mine-wide shutdowns
- 2 Extend duration between bi-weekly shuts
- 3 Increase working hours on mining machines
- 4 Improve short-term planning SIC
- 5 Improve mill operational availability
- 6 Increase hoist productivity
- 7 Improve conveyor operational availability
- 8 Increase on line ore cutting (process)
- 9 Increase on line ore cutting (technology)
- 10 Ensure mine planning aligned across system
- 11 Ore separation UG
- 12 Increase capacity of mill feed stockpile
- 13 Improve solid bowl reliability
- 14 Optimal feed rate when running
- 15 Minimize fines generation
- 16 Cross-feed Hoists to Mills
- 17 Reduce breakdown maintenance and excessive wear
- 18 Improve maintenance productivity
- 19 Optimize reagent consumption
- 20 Achieve target product concentration
- 21 Overtime usage in mine production
- 22 Energy management (site wide)
- 22 Loadout automation

Our offices are currently based in 3 countries, and we operate globally



The EGCA leadership team has deep experience in reducing operating costs for mining clients



Odilon Kongolo
Chief Executive Officer

- > 15 Years Experienced consultant and registered professional nuclear structural engineer
- Delivered production ramp-up, cost reduction, and capital optimization across multiple mining operations, achieving ~\$100m savings, R20.5bn NPV uplift, and R300m pipeline execution through operational, system, and project excellence
- Expert at digital twin implementation



Morkel Erasmus
Managing Director: RSA

- Industrial engineer with 20 years' experience in petrochemical, mining, supply chain operations and management consulting
- Scoped, led and delivered multiple cost optimization projects in mining across Africa, delivering > \$100m in annual benefits for clients
- Deep experience in rolling out continuous improvement programs, as well as embedding management operating systems, cost management policies and supplier management strategies in mining across Africa



Patrick Kambala
Managing Director: DRC

- Management consultant with over 20 years of mining experience across open-pit, underground, and cement operations
- Led greenfield projects, optimized production and cost efficiency, implemented grade control and HSE systems, and improved operational performance
- Proven ability to align mining strategy with business goals, driving sustainable value, safety, and profitability across diverse mining environments.

We have worked with various large mining and industrial clients
and achieved sustainable results



GLENCORE



ANGLO
AMERICAN



sasol





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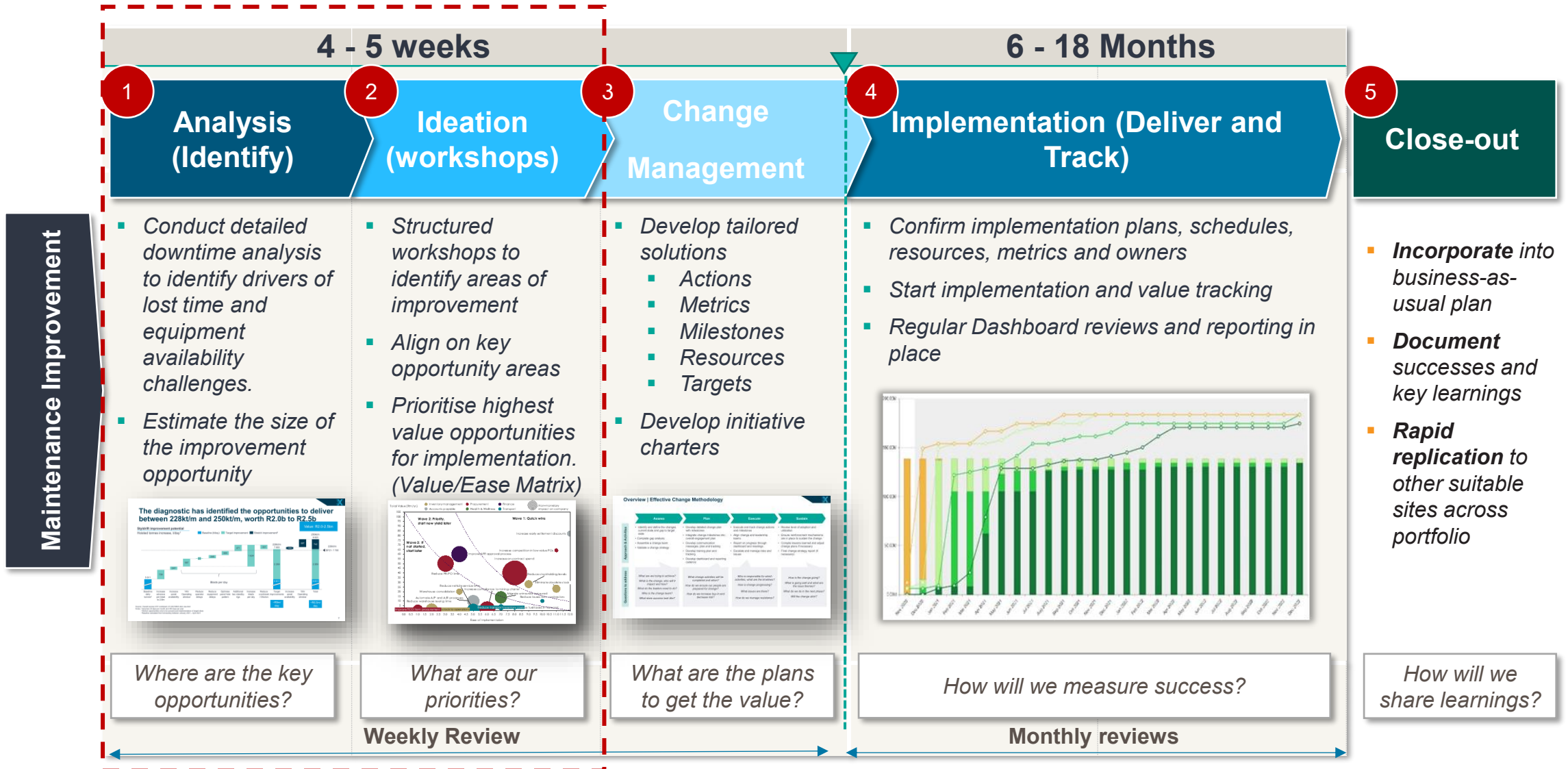
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How we think about improving availability

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Proposed next steps

Our approach follows 5 non-negotiable steps which are key to sustaining results.



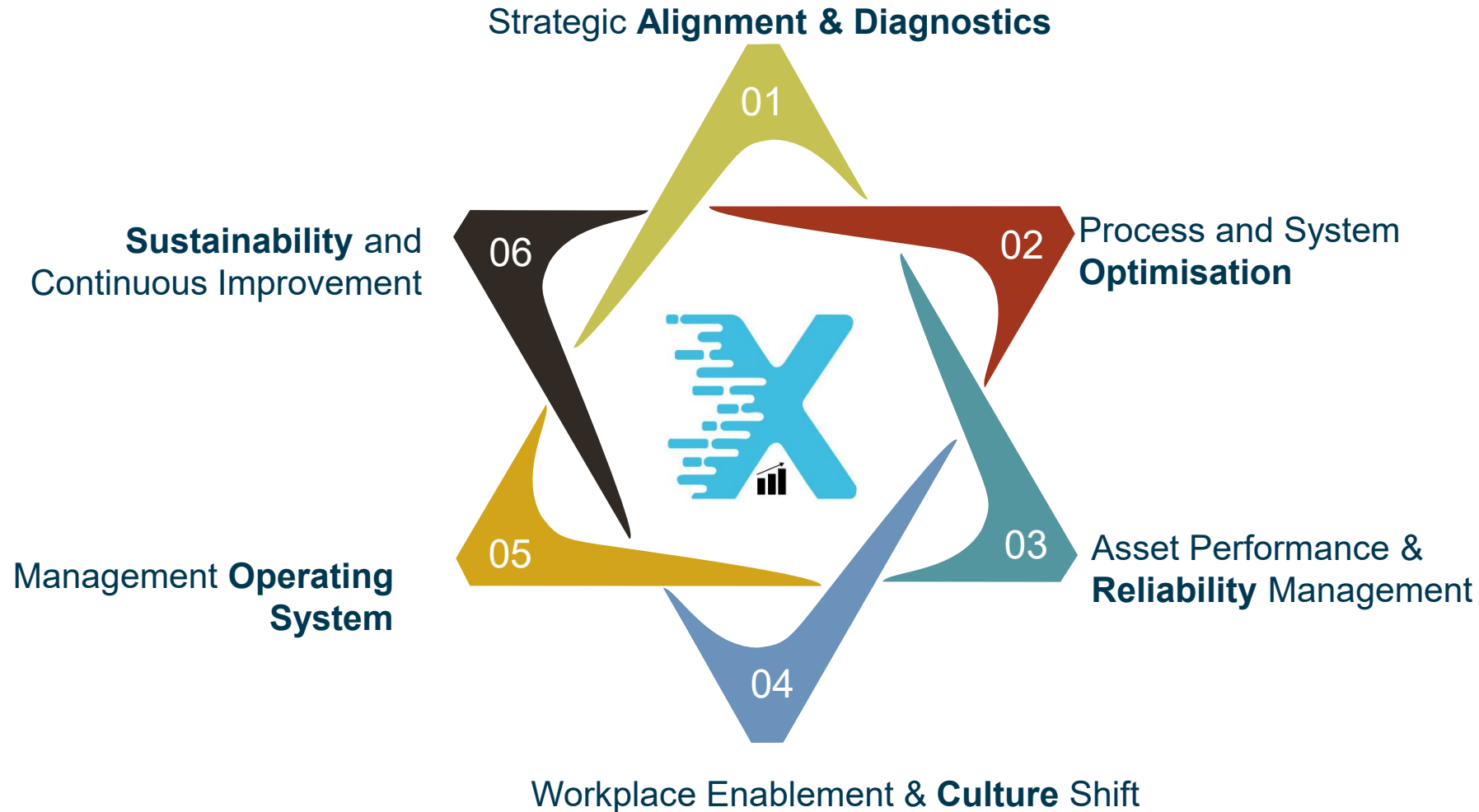
...the implementation and tracking continues until full value of initiatives is realised.



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Our implementation approach prioritises long-term sustainability and revolves around 6 essential inter-linked activities.



Our methodology is **not linear**—it's **adaptive and iterative**, ensuring that lessons learned in one arm are fed into others in real time. Change management and communication are woven through all six arms to ensure adoption and sustainability.

Strategic Alignment & Diagnostic: Ensure all improvement efforts directly support the organization's strategic objectives

Key activities:

1

Executive Alignment Workshops



2

Business diagnostics (financial, operational, people)



3

Baseline performance benchmarking



4

Identification of high-impact value levers



5

Target setting with a clear business case

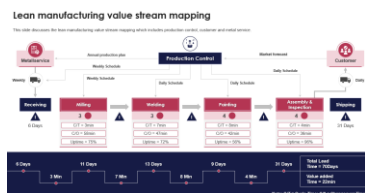


2 Process & Systems Optimization: Identify and remove inefficiencies through lean, digitised, and standardised processes

➤ Key activities:

1

Process mapping and value stream analysis



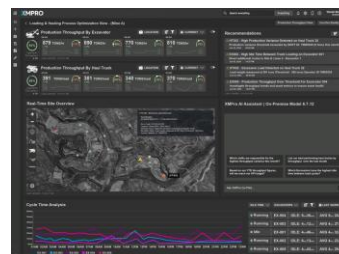
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Optimizationn (Value Engineering)



3

Workflow automation and system integration



4

Policy and procedure refinement



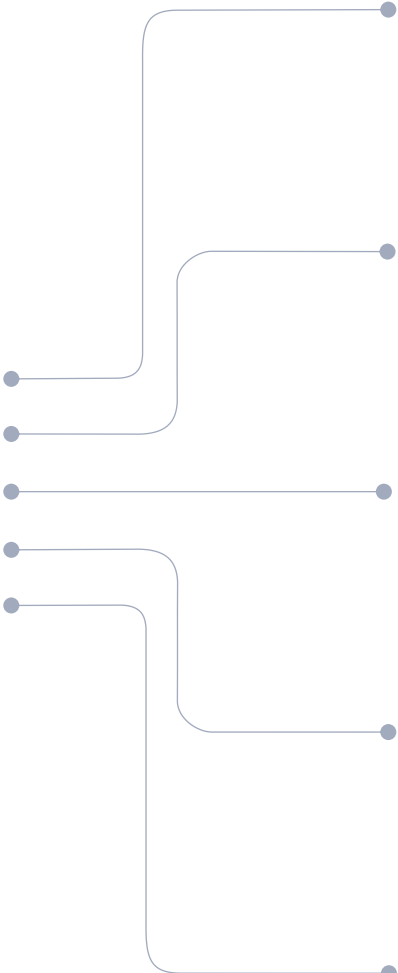
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Elimination of waste and non-value-adding tasks



Asset Performance & Reliability Management: Maximize availability, reliability, and return on capital assets


Key activities:



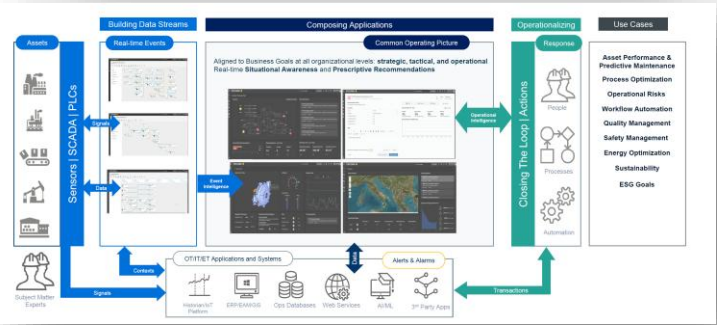
**Critical asset strategy review**

**Reliability-centered maintenance (RCM)**

**Condition monitoring and predictive maintenance**

**Asset hierarchy & master data clean-up**

**Integration of iot/digital twin technologies**



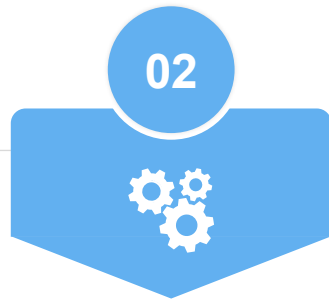


5 Management Operating System (MOS) & Performance Rhythm: Institutionalise effective decision-making, visibility, and accountability at all levels.

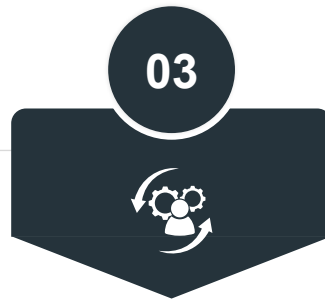
➤ Key activities:



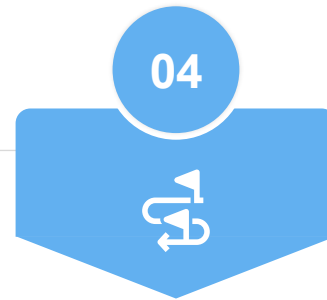
Design and implementation
of MOS (tiered meetings,
KPIs, dashboards)



Daily/weekly/monthly
performance reviews



Visual management
and action tracking
tools



Leader standard work
and escalation
mechanisms



Integration of digital
tools for performance
management

6 Sustainability & Continuous Improvement Engine: Ensure improvements are locked in and continuously enhanced after implementation

➤ Key activities:

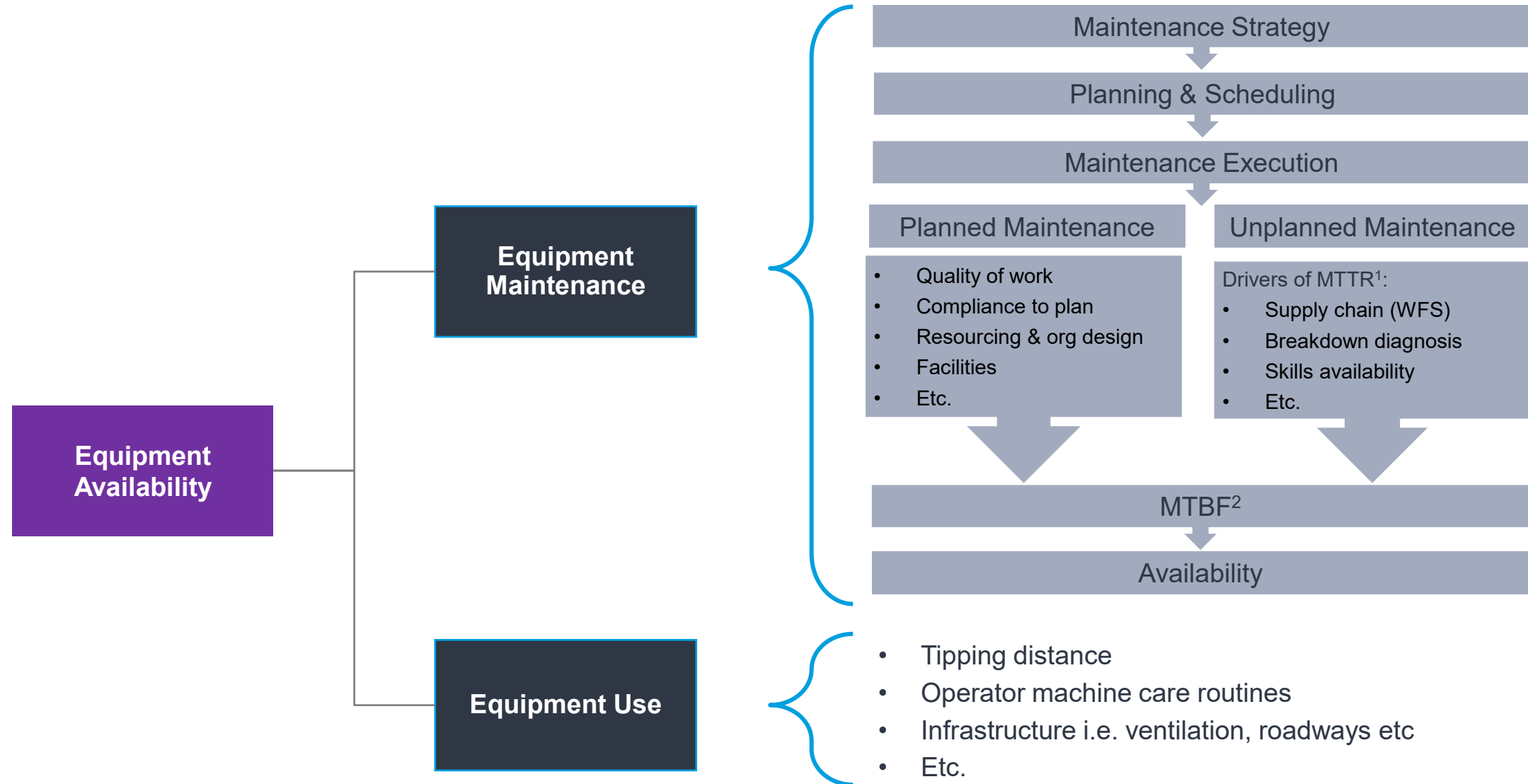




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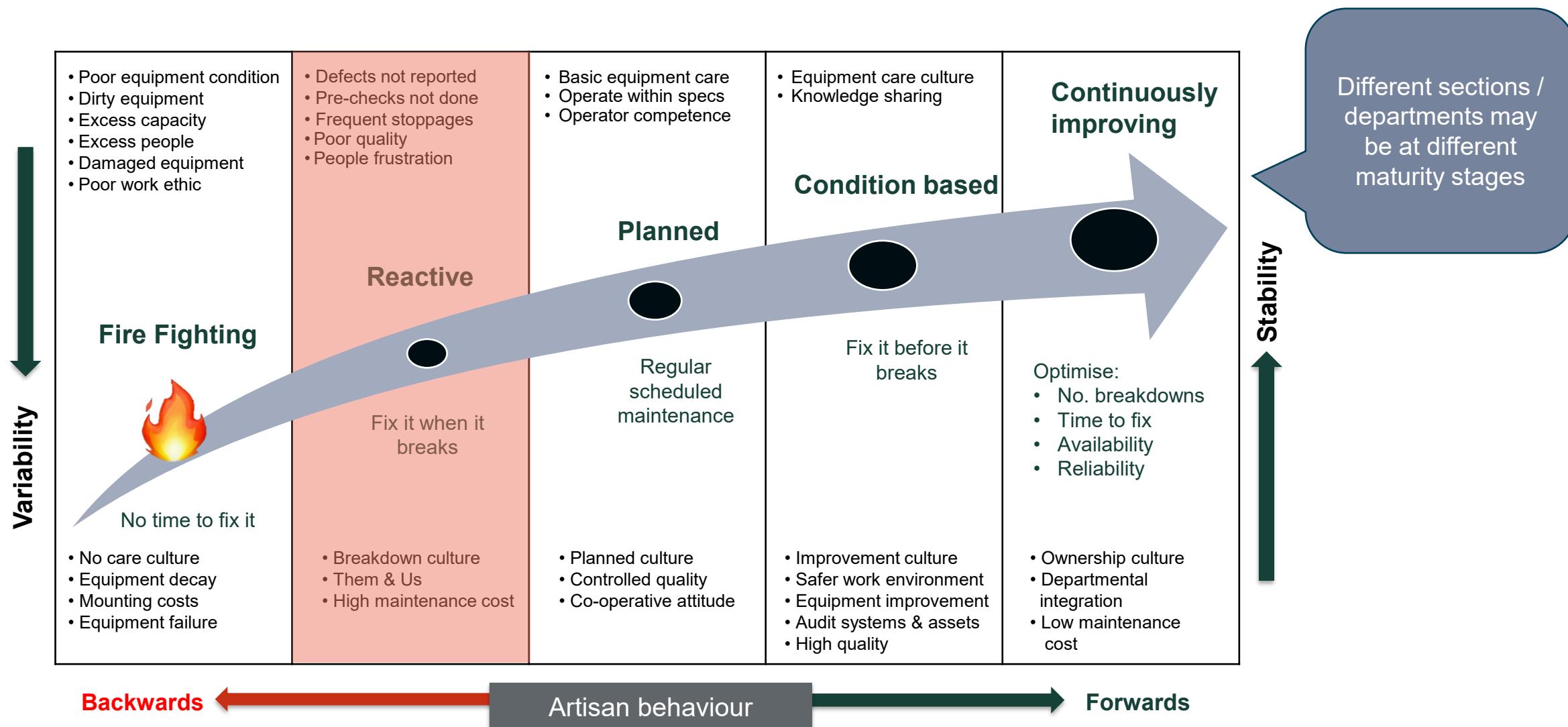
Our diagnostic process will assess the drivers of plant and equipment availability and identify the priority opportunities



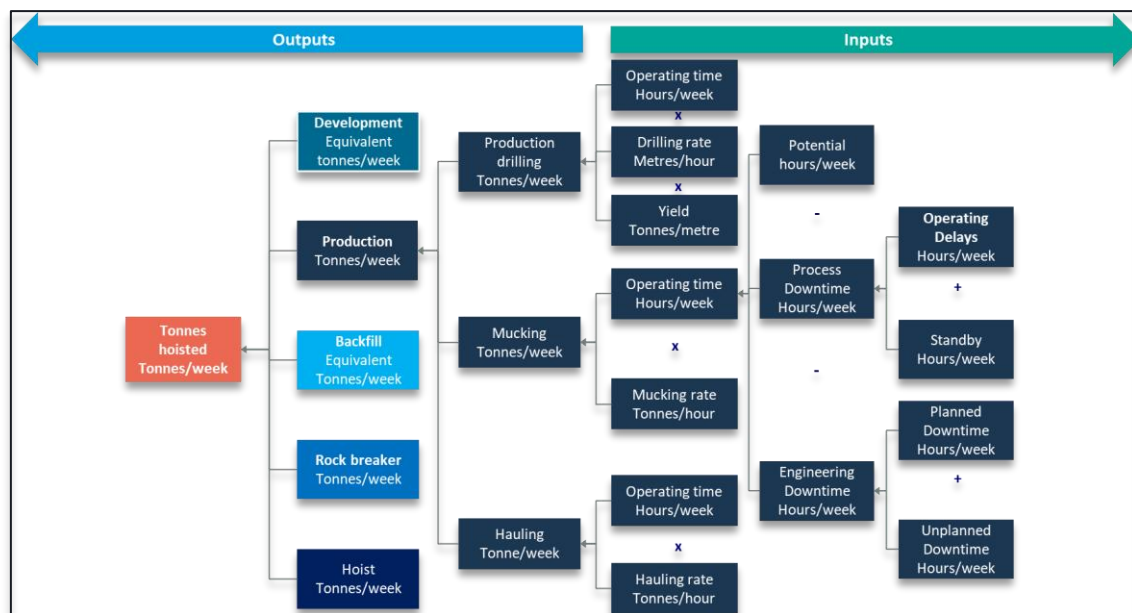
1. MTTR: Mean Time to Repair

2. MTBF: Mean Time Between Failure

We will assess the maintenance maturity of ARM Bokoni to determine the path to away from reactive maintenance



We know from experience which levers drive improvement in availability



Value driver trees (VDTs) linked to the marginal value of extra tonnes/availability

Hypothesis-driven approach to solving the problem, across engineering, supply chain and operations departments

Improvement levers to explore		Equipment Availability		Illustrative
		Frequency (MTBF)	Duration (MTTR)	
Maintenance strategies (incl. inspections and condition monitoring)		- Component strategies run to failure and PMs don't incl all required activities - Inspections /condition monitoring does not identify corrective actions - Poor data capturing (incl. poor data hierarchy) and lack of automated reporting stops teams from efficiently and effectively understanding equip. performance - Gaps in RCA and improvement process limit ability to drive equipment performance and improve the maintenance system		
Planning and scheduling		- No work management KPIs (e.g. backlog, schedule loading, critical parts levels, PM and schedule compliance) means people can not be held to account - Material gaps in works management MOS and review cascade means the incorrect work is prioritised and not completed on time - No tools to help make planning and scheduling easy to do and interpret (e.g. Prometheus and or simple excel ones)		
Materials management		Low quality parts (e.g. cylinders) used (not right contract /not on contract)	- Critical parts /consumables out of stock when req (not defined or tracked) - Parts not available at work site due to poor underground stores / logistics	
Execution		- Training and task lists do not support effective, quality work on key tasks - Insufficient over inspections / QA for key components and inspections	- Delays due to travel time to and from remote workshops - Resources not available for repairs / logistics (absenteeism / short shifting) - Equipment not commission when repairs done (e.g. turbos, cylinders)	
Organisation / Capability		- Front line teams do not know priorities for the day and how their section is performing - KPIs and visual boards not built down to front line - Engineering does not have sufficient to identify, plan and drive improvements (currently on 1 person holding ticket, doing projects and reliability) - Planning does not have required capacity and status to ensure work correctly prioritized and done timeously - Maintenance execution structure does not create required single point accountability for section performance and planned / key corrective work - Foreman do not have knowledge to drive maintenance system and artisans do not have required technical training on key equipment (e.g. new engines)		

The goal is to map and prioritise each input lever to the opportunity linked to improved availability

Illustrative

Potential downtime reduction for LHDs

% availability, improvement vs baseline

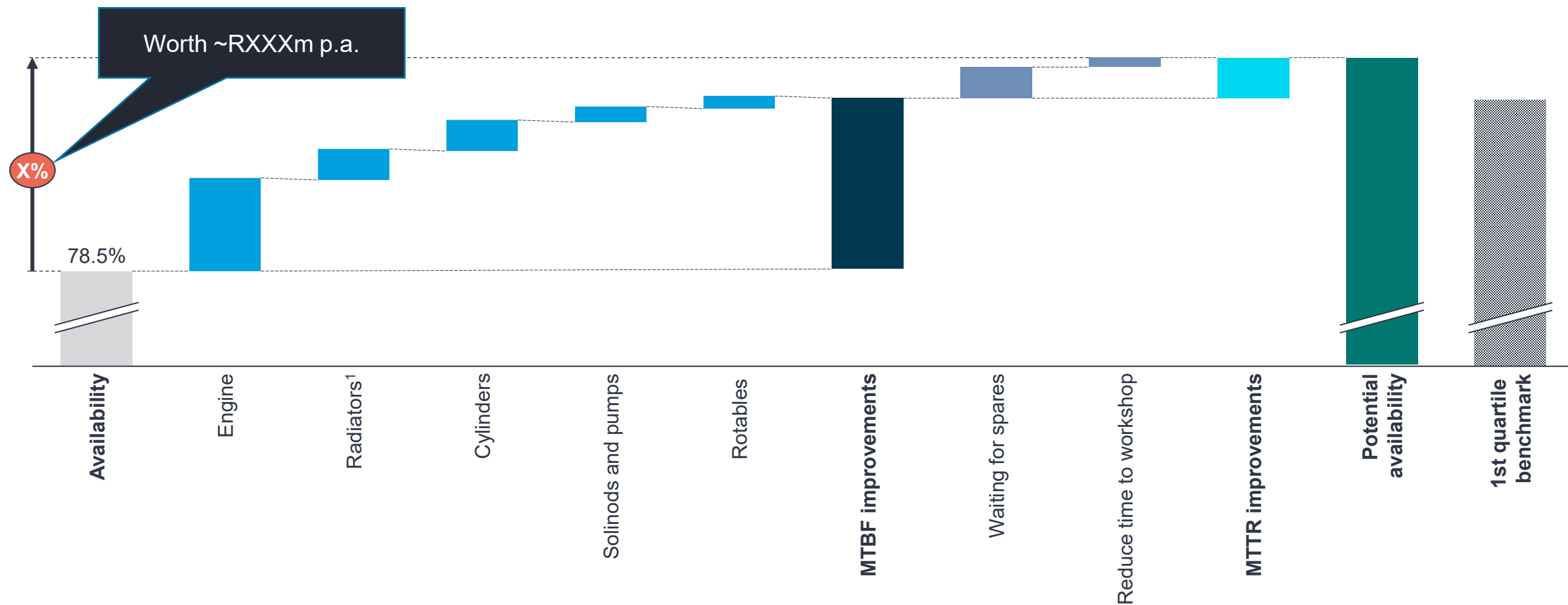




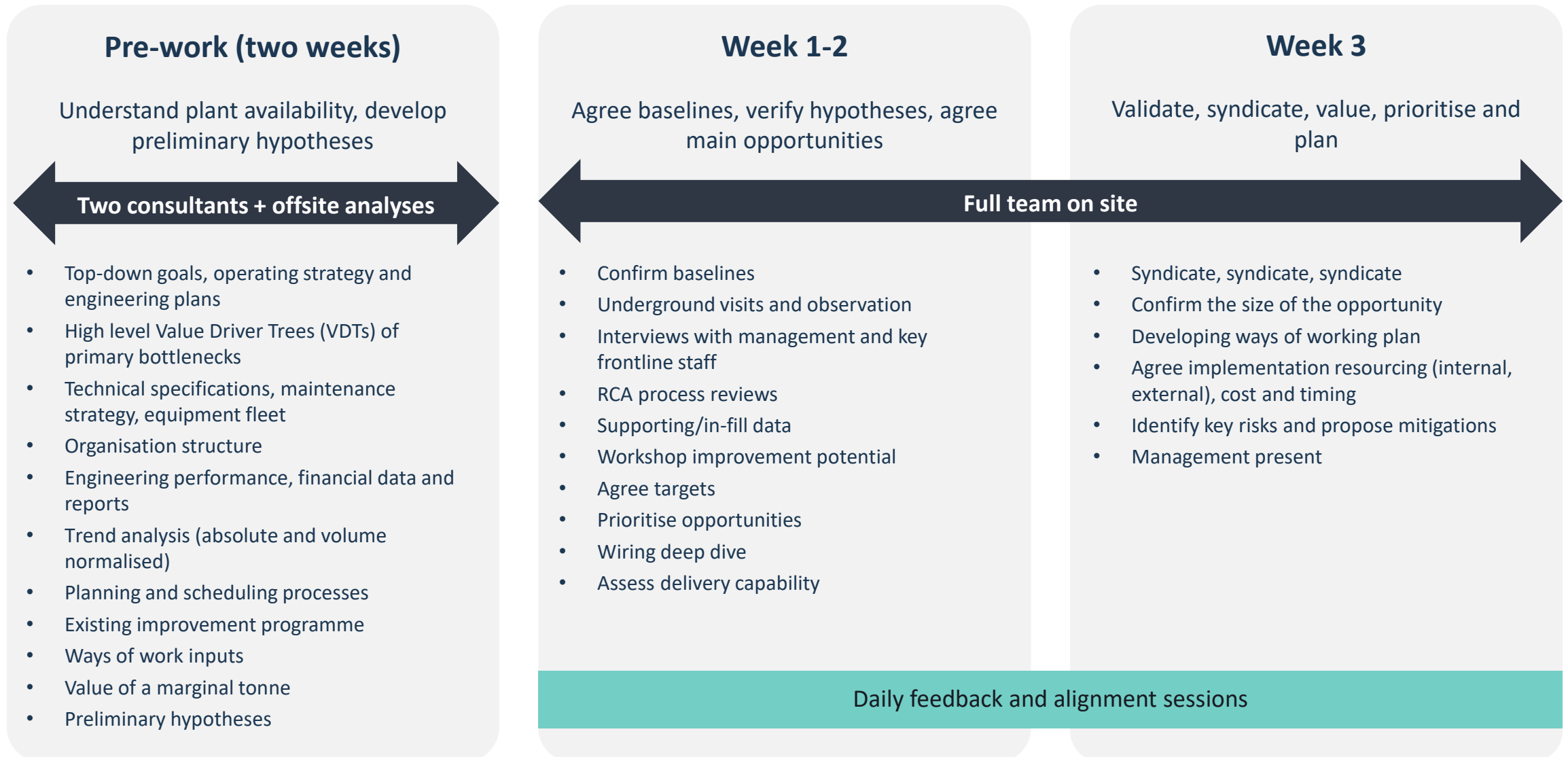
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Next steps

- We propose a maintenance improvement diagnostic for ARM Bokoni which will span 6 weeks (2 weeks pre-work, followed by 4 weeks on-site)
- The output of the diagnostic will be a joint understanding of the nuanced challenges across various departments/sections, and a detailed presentation outlining the size of the opportunity, with a workplan outlining the work and resourcing required to unlock the opportunity.

The diagnostic focuses on on-site observations, stakeholder interaction, verification of hypotheses, and syndication





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